

No. of Functional Features Included in the Solution

Date	02 July 2026
Team ID	XXXXXX
Project Name	A Comprehensive Measure of Well-Being
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	HDI Score Prediction	Predicts the HDI score using four socioeconomic indicators through a Linear Regression model.	Flask (/predict)	Done	High
2	Tier Classification	Classifies the predicted HDI as Low, Medium, High, or Very High.	classify_hdi()	Done	High
3	Input Validation	Validates all input values before prediction and displays appropriate error messages.	validate_inputs()	Done	High
4	Flask Web Interface	Provides Home, Prediction, and About pages for user interaction.	Flask Templates	Done	High
5	EDA Visualizations	Generates heatmaps, scatter plots, pair plots, and distribution plots for analysis.	Jupyter Notebook	Done	Medium
6	Model Serialization	Saves and loads the trained Linear Regression model using Pickle (HDI.pkl).	Pickle	Done	Medium
7	About Page	Displays project information, methodology, and HDI classification details.	about.html	Done	Low
8	Error Handling	Handles invalid URLs and incorrect user inputs gracefully.	Flask Error Handler	Done	Low

Feature Summary:

Item	Details
Total Features Planned	8
Total Features Implemented	8

Core / Must-Have Features	HDI Prediction, Input Validation, Tier Classification, Flask Web Interface
Additional Features	EDA Visualizations, Model Serialization, About Page, Error Handling
Features Tested & Verified	All 8 features successfully tested

Feature Category Breakdown:

S.No	Category	Features in Category
1	User Interface (UI)	Home page, Prediction page, About page, Result display, Error page
2	Backend / Logic	Flask routing, Input validation, HDI prediction, Tier classification
3	Data Storage	HDI.csv dataset and HDI.pkl trained model
4	Machine Learning	Linear Regression model training, prediction, and evaluation
5	Security & Validation	Input validation, range checking, and safe handling of invalid inputs